QFT

- usually starts from continuum classical field theory (such as E&M) by quantizing its normal modes (more technically, by replacing Poisson brackets with appropriate commutation relations)

  → this is essence of Quantum Electrodynamics (QED)

- There are many other field theories that emerge in various condensed matter situations, in particular in the vicinity of <u>continuous phase transitions</u>

  → strange degrees of freedom can emerge from very simple models (fermions in transverse field Ising chain)

- Fermions interacting with various collective fields is the essence of all modern field theories (plasma and superconductivity are such two key examples)

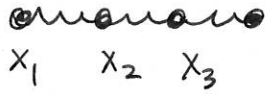
$$\left.\begin{array}{l}\text{Universality}\\ \text{Symmetry}\end{array}\right\}\ \text{emerge at low energies in many different physical systems}$$

QFT: (1) quantum mechanical models of subatomic particles in particle physics

(2) quasi-particles in condensed matter physics

Classical fields

Harmonic chain = chain of coupled oscillators. ~~Position~~ ~~$R_n = na + x_n$~~



each particle is characterized by coordinate x_n and momentum p_n

Classical Hamiltonian $H = \sum_n \frac{p_n^2}{2m} + \frac{1}{2} K_0 (x_{n+1} - x_n)^2$

$$\frac{dx_n}{dt} = \frac{\partial H}{\partial p_n} \Rightarrow \dot{x}_n = \frac{p_n}{m}$$

$$\frac{dp_n}{dt} = -\frac{\partial H}{\partial x_n} \Rightarrow -\dot{p}_n = K_0 (x_{n+1} - x_n)(-1) + K(x_n - x_{n-1})$$

$$= K_0 (2x_n - x_{n+1} - x_{n-1})$$

$$\Rightarrow \ddot{x}_n = \frac{1}{m}\dot{p}_n = \frac{K_0}{m}(x_{n+1} - 2x_n + x_{n-1}) \qquad (1)$$

$$\ddot{p}_n = -K_0(2\dot{x}_n - \dot{x}_{n+1} - \dot{x}_{n-1}) = \frac{K_0}{m}(p_{n+1} - 2p_n + p_{n-1}) \qquad (2)$$

Try plane wave solution

$$x_n = u_k e^{i(kn - \omega_k t)} + u_k^* e^{-i(kn - i\omega_k t)}$$

Note: $x_n \in$ real, while $u_k \in$ complex.

$$\ddot{x}_n = -\omega_k^2 x_n = \frac{K_0}{m}\Big\{ u_k e^{-i\omega_k t}\left(e^{ik(n+1)} + e^{ik(n-1)} - 2e^{ikn}\right)$$

$$+ u_k^* e^{i\omega_k t} e^{-ikn}\left(e^{-ik} + e^{ik} - 2\right)\Big\}$$

$$\Rightarrow -\omega_k^2\left(u_k e^{i(kn - \omega_k t)} + u_k^* e^{-i(kn - \omega_k t)}\right) =$$

$$= \frac{2K_0}{m}(\cos k - 1)\left(u_k e^{i(kn - \omega_k t)} + u_k^* e^{-i(kn - \omega_k t)}\right)$$

$$\omega_k^2 = \frac{2K_0}{m}\underbrace{(1-\cos k)}_{2\sin^2\frac{k}{2}} = \frac{4K_0}{m}\sin^2\left(\frac{k}{2}\right)$$

$$\omega_k = 2\sqrt{\frac{K_0}{m}}\left|\sin\frac{k}{2}\right| \approx \sqrt{\frac{K_0}{m}}\,k$$

$$\Rightarrow \quad x_n(t) = \sum_k \left(u_k\, e^{i(kn-\omega_k t)} + u_k^*\, e^{-i(kn-\omega_k t)}\right)$$

mode expansion : u_k, u_k^* complex amplitude

$$\omega_k = 2\sqrt{\frac{K_0}{m}}\left|\sin\frac{k}{2}\right|$$

peridic function of k $\Rightarrow$ restrict $k \in (0, 2\pi)$

or $k \in (-\pi, \pi)$.

Periodic boundary condition: $x_{n+N} = x_n$

(N-particle chain)

$$\Rightarrow e^{ikN} = 1$$

$k_\ell = \frac{2\pi \ell}{N}$ with $\ell \in \left(-\frac{N}{2}, \frac{N}{2}\right)$

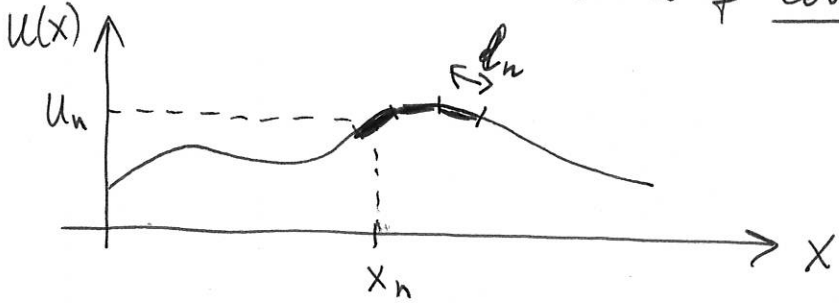
takes N values.

Thus, with x_n, $n \in (1, 2, 3 \ldots N) \Rightarrow N$ values

Amplitudes u_k also take N different values,

one for each allowed k_ℓ.

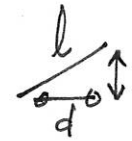
<u>Field description</u>: instead of discrete atoms, think of <u>continuous</u> elastic string



String is "made" of segments of length d, each of mass ρd (ρ = linear mass density)

$\Rightarrow$ for the <u>n-th</u> segment $p_n = \rho d \frac{\partial u_n}{\partial t}$

$$K.E. \Rightarrow \sum_n \frac{1}{2} \frac{p_n^2}{\rho d}$$



$l^2 = d^2 + (\Delta u)^2$

deformation

$$P.E. \Rightarrow \sum_n \frac{1}{2} \varkappa l_n^2 =$$

$$= \frac{1}{2} \varkappa \sum_n \left(d^2 + (\Delta u_n)^2 \right)$$

Can drop $\sum_n \left(\frac{1}{2} \varkappa d^2 \right)$ as it describes ~~[illegible]~~ non-deformed string

$$H = \sum_n \frac{p_n^2}{2\rho d} + \frac{1}{2} \varkappa \left(u_{n+1} - u_n \right)^2 =$$

$$= \sum_n d \left(\frac{p_n^2}{2\rho d^2} + \frac{1}{2} \varkappa d \left(\frac{u_{n+1} - u_n}{d} \right)^2 \right)$$

$\Delta u_n \approx u_{n+1} - u_n$ as long as $\Delta u_n \ll d$

<u>Continuum limit</u> $d \to 0$

$$\underbrace{\frac{u_{n+1} - u_n}{d}}_{\frac{u(n+1)d - u(nd)}{d} = \frac{u(x+\Delta x) - u(x)}{\Delta x}} \to \frac{\partial u}{\partial x}, \quad \frac{p_n}{d} \to p(x) \text{ (momentum density)}$$

$$H = \sum_n \underbrace{d}_{\Delta x} \underbrace{\{\quad\}}_{\text{Hamiltonian density } \mathcal{H}} \longrightarrow \int dx\, \mathcal{H}(x)$$

$$\mathcal{H} = \frac{p(x)^2}{2\rho} + \frac{1}{2} K_0 \left(\frac{\partial u(x)}{\partial x}\right)^2.$$

where $K_0 = \text{xed}$ stays finite in $d \to 0$ limit.

Thus $H = \int dx \left\{ \frac{p^2}{2\rho} + \frac{1}{2} K_0 \left(\frac{\partial u}{\partial x}\right)^2 \right\}$

Equation of motion: $\frac{\partial H}{\partial p_n}$, $\frac{\partial H}{\partial x_n}$ are to be replaced by variational derivatives $\frac{\delta H}{\delta p(x)}$, $\frac{\delta H}{\delta u(x)}$

$$\begin{cases} \dfrac{\partial u(x,t)}{\partial t} = \dfrac{p(x,t)}{\rho} \\[2ex] \dfrac{\partial p(x,t)}{\partial t} = -\dfrac{\delta H}{\delta u} = K_0 \dfrac{\partial^2 u}{\partial x^2} \end{cases}$$

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = \dfrac{1}{\rho} \dfrac{\partial p}{\partial t} = \dfrac{K_0}{\rho} \dfrac{\partial^2 u}{\partial x^2} \\[2ex] \dfrac{\partial^2 p}{\partial t^2} = K_0 \dfrac{\partial^2}{\partial x^2} \dfrac{\partial u}{\partial t} = \dfrac{K_0}{\rho} \dfrac{\partial^2 p}{\partial x^2} \end{cases}$$

wave eq.

Lagrangian $L = p\dot{q} - H$

Guess that $L = \int dx \left\{ \frac{1}{2} \rho \left(\frac{\partial u}{\partial t}\right)^2 - \frac{1}{2} K_0 \left(\frac{\partial u}{\partial x}\right)^2 \right\}$

$$\frac{\delta L}{\delta u} - \frac{d}{dt} \frac{\delta L}{\partial \dot{u}} = 0 \Rightarrow K_0 \frac{\partial^2 u}{\partial x^2} = \frac{d}{dt} \rho \dot{u} = \rho \frac{\partial^2 u}{\partial t^2}$$

$\Rightarrow$ get the same wave eq for $u(x,t)$ back.